

“Nesting assistance”: a new hypothesis for the evolution of polyandry and a test in an African foam-nesting treefrog

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Abstract

Why polyandry evolves remains a compelling evolutionary question. While numerous material benefits of polyandry have been demonstrated, there has been limited attempt to link the fitness of polyandrous females to nesting behavior. Here, we introduce the “nesting-assistance” hypothesis as a new explanation for the evolution of polyandry. We theorize that in species where females are primarily responsible for nest construction, and males display some form of nesting assistance, that mating with multiple males leads to the production of more viable nests and improved offspring survival or performance. We provide an initial test of the “nesting-assistance” hypothesis in the African gray foam-nest treefrog (*Chiromantis xerampelina*), a species where multiple males commonly join females to construct arboreal foam nests. Field observations revealed that individuals involved in polyandrous matings synchronize the timing of bouts of leg churning, a conspicuous reproductive behavior essential for foam-nest construction. Moreover, matings involving more males produced bigger nests that were more resistant to desiccation. Paternity analysis revealed that males involved in polyandrous matings gained a share of paternity, suggesting a direct fitness benefit from nesting assistance. Our findings support the “nesting assistance” hypothesis and provide evidence for a new yet potentially widespread benefit of polyandry.

Keywords: mating system, polyandry, amphibian, sexual selection, nesting

Introduction

Polyandry was once thought to be rare in nature but is now known to be taxonomically widespread. Initially, polyandry was thought to be the outcome of males forcing copulation against female interests (sexual coercion) or females accepting superfluous matings to minimize costs of resistance (convenience polyandry) (Parker & Birkhead, 2013). Consideration has also been given to the possibility that polyandry is an artifact of females reducing the risk of missed-mating opportunities. In a null model of polyandry, females might accept matings with a series of males to reduce the risk of failing to encounter an acceptable male and remaining unmated (Kokko & Mappes, 2013). However, progressive theoretical developments, combined with detailed observational and empirical studies, have collectively shown that polyandry can be an adaptive female mating strategy. Females stand to benefit directly if polyandry secures extra nutrients for progeny (either through seminal fluid proteins or by exchanging matings for food) (Arnqvist & Nilsson, 2000), reduces the risk of infanticide (Klemme & Ylönen, 2010), or safeguards against pairing with a male who is either a poor father (Kempnaers, 1993) or has low fertility (Sheldon, 1994). Polyandrous females can also benefit indirectly by elevating offspring genetic diversity and adaptability (Jennions & Petrie, 2000), reducing inbreeding depres-

sion (Cornell & Tregenza, 2007), or improving offspring genetic quality and viability, or attractiveness (Slatyer et al., 2012). While these benefits are truly diverse, we can expect that additional benefits will be uncovered as we more closely examine female mating behavior, especially in systems where males invest heavily in reproduction through the provision of mate assistance.

One overlooked mechanism by which polyandrous females might elevate their fitness is linked to nesting strategy. Nesting involves parents building or modifying structures or sites to house their progeny and primarily functions to protect developing offspring from environmental factors that threaten survival. Here, we introduce a new concept that links polyandry with nesting behavior and male assistance: the “nesting-assistance” hypothesis, in which improved nesting outcomes favor the evolution of polyandry in species where reproductive success is strongly tied to nesting success. The “nesting assistance” hypothesis predicts that in species where females are primarily responsible for constructing nests, and males either directly or indirectly assist with nest construction, that mating with multiple males leads to the production of more viable nests and improved offspring survival and/or performance. A primary assumption of the hypothesis is that assisting males will elevate their fitness by gaining a share of paternity. We propose that

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the hypothesis will be (i) most applicable to species where there is a male-biased operational sex ratio (OSR), creating a situation where multiple males are available to assist individual females during nest construction, and (ii) applicable to species with either external or internal fertilization, although restricted to contexts where all participating males have at least some opportunity to gain fertilizations and reproductive success. On this basis, the hypothesis may have limited or no application to species with internal fertilization when there is complete first or last male sperm precedence because such systems will preclude multiple paternity and male-fitness rewards for helping.

Across taxa, male nesting assistance and multiple paternity have been reported in a diversity of vertebrate and invertebrate taxa, including New World monkeys (e.g., tamarins and marmosets: Goldizen, 1988; Smith et al., 2007), canids (e.g., arctic foxes and gray wolves: Ausband, 2018; Carmichael et al., 2007), raptors (e.g., Harris hawks and Galapagos hawks: DeLay et al., 1996; Faaborg et al., 1995; Mader, 1979), seabirds (e.g., common terns and Red-footed Booby: Cao et al., 2010; Ludwigs, 2004), waterfowl (e.g., dusky moorhens and Tasmamien native hens: Garnett, 1980; Gibbs et al., 1994; Goldizen et al., 1998; Putland & Goldizen, 2001), cichlid fishes (e.g., arapaima fish and cave-spawning cichlid fish: Awata et al., 2005; Farias et al., 2015), monitor lizards (e.g., Rosenberg's goanna: Rismiller et al., 2010), solitary bees (e.g., carpenter bees: Mikát et al., 2019; Mikát et al., 2021) and solitary wasps (e.g., organ-pipe mud-dauber wasps: Brockmann, 1992). While these systems provide considerable opportunity to test the "nesting-assistance" hypothesis, another group where polyandrous females might strongly benefit from securing assistance from multiple males during nest construction are anuran amphibians. Nesting in anurans has its origins in the evolutionary transition from water to land and increased selective pressure to maintain an aquatic environment that protects eggs and larvae from water loss and desiccation (Fischer, 2023). Anurans display a variety of nesting behaviors, but one of the most conspicuous is the construction of foam nests. Typically, foam nests are built above water or on land (including subterranean and arboreal sites) with nest construction initiated by females (Gould, 2021). After choosing a nest site, females pair with a male and release oviducal secretions derived from egg jelly (along with water expelled from the bladder) (Gould, 2021). Rhythmic churning of the arms or legs (displayed by the female, the male, or both sexes) is then used to mechanically agitate and aerate the viscous blend to form a spherical or ellipsoid foam structure (Heyer & Rand, 1977). The inclusion of air allows the nest to inflate to a much greater size (Gould, 2021), and pairs that invest more time in mechanical agitation can potentially produce larger nests (Dalgetty & Kennedy, 2010). During nest construction, eggs are typically released in discrete deposition bouts (accompanied by bouts of arm or leg churning), and then externally fertilized and integrated into the growing nest. Following nest completion, drying of the outer foam layer creates a hard shell that traps moisture until eggs hatch and larvae break free (Fischer, 2023; Gould, 2021). The size and integrity of the foam nest appear to be critical for success, with evidence that smaller nests suffer higher rates of evaporative water loss and egg mortality (Gould, 2021; Seymour & Loveridge, 1994; Zina, 2006) and that disruption to the nest shell can quickly result in egg desicca-

tion (Shepard & Caldwell, 2005). With such a high risk of complete brood failure, it is conceivable that improved nest integrity gained through multiple males assisting with nest construction could directly favor the evolution of polyandry.

In various foam-nesting frog species, multiple males have been observed to join females during nest construction, and it is widely assumed that males involved in polyandrous matings provide nesting assistance by churning their legs in a coordinated fashion, yielding fitness benefits to those involved (Coe, 1967; Heyer & Rand, 1977; Prado & Haddad, 2003). In support of this notion, precopulatory physical competition between males is rarely observed, and there appears to be some level of synchrony in limb-churning behaviors essential for foam production (Coe, 1967; Fukuyama, 1991; Heyer & Rand, 1977; Hödl, 1992; Liao & Lu, 2010; Prado & Haddad, 2003), although this remains to be quantified. Moreover, parentage analysis has shown that participating males can be rewarded with a high share of paternity, even from very peripheral spawn positions (Zhao et al., 2016). For foam-nesting treefrogs from the African genus *Chiromantis*, polyandrous mating is especially prominent, with conspicuous arboreal foam nests constructed on vegetation overhanging water and multiple male involvement with nesting described in all four species (*C. rufescens*, *C. kelleri*, *C. petersii*, and *C. xerampelina*) (Coe 1967; Taylor 1971; Coe 1974; Jennions et al. 1992; Byrne & Whiting, 2008). Coe (1967) closely observed the nesting process in the Western foam-nest treefrog (*C. rufescens*) and speculated that coordinated leg churning among builders might lead to the production of larger nests that facilitate the rearing of larger clutches. Beyond the potential to accommodate larger clutches, we hypothesize that male involvement in foam nest construction may lead to higher levels of mechanical agitation and aeration and generate larger nests that are more resistant to desiccation, improving offspring survival and elevating female fitness.

The aim of this paper is to introduce the "nesting-assistance" hypothesis for the evolution of polyandry and provide a first test of this hypothesis in the African gray foam-nest treefrog, *C. xerampelina*. To test our key predictions that males provide nesting assistance, that this assistance increases nest viability, and that males gain a fitness benefit from assisting, our specific objective was to answer four questions. First, do frogs synchronize bouts of leg-churning behavior during nest construction? If males are directly or indirectly assisting females to build nests through the simultaneous and rhythmic movement of hind limbs needed to mechanically agitate foam, we should expect that males will synchronize their leg movements with bouts of female leg churning (and egg release). Second, do matings involving more males have a higher proportion of male-leg churning activity during nest-building bouts? If polyandrous females receive more male-nesting assistance, we should expect that the proportion of time during a leg-churning bout that involves at least one male churning in synchrony with the female will increase with the number of participating males. Third, are nests with more males larger and less susceptible to desiccation? If male-leg churning is assisting females to produce more viable nests, we should expect a significant positive relationship between the number of males involved in nest construction and nest size, and that larger nests will be less susceptible to desiccation. Fourth, do males involved in polyandrous nests receive a fitness benefit

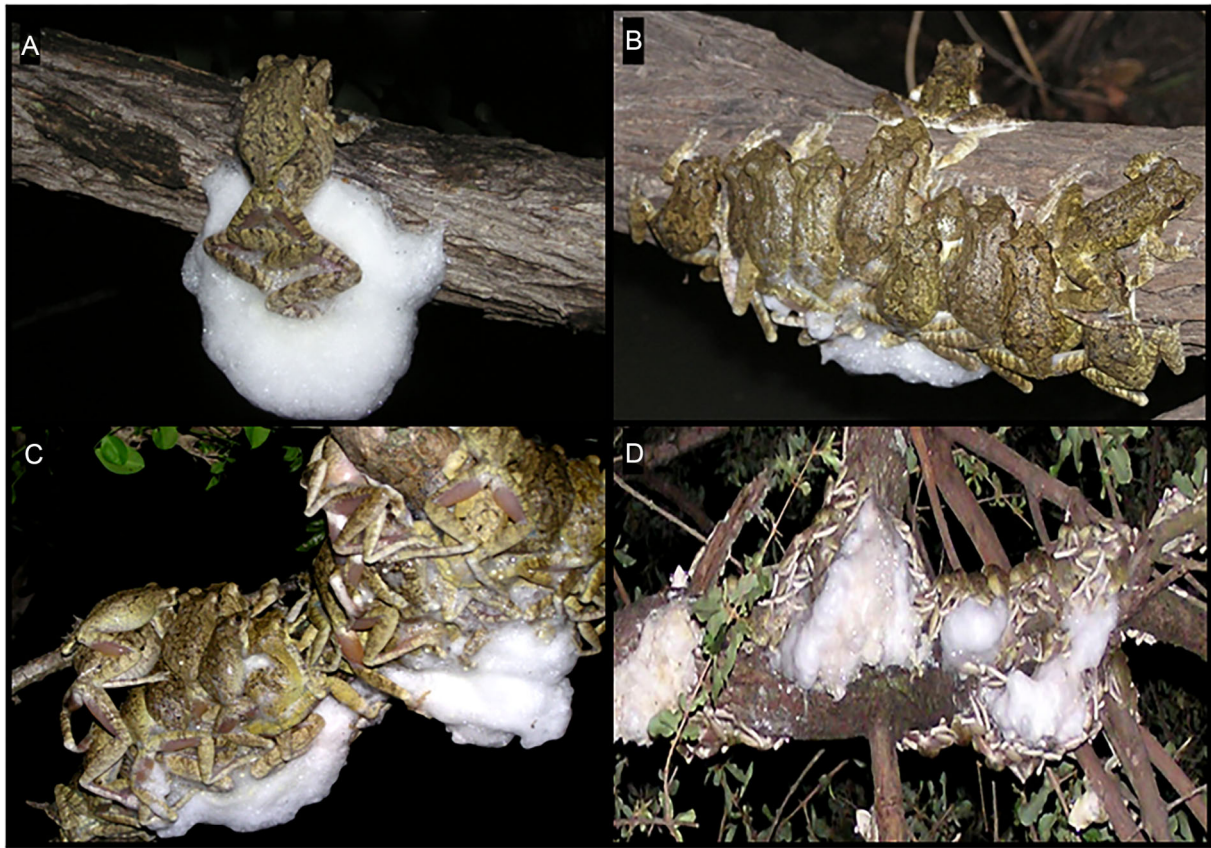


Figure 1. African foam-nest treefrogs (*Chiromantis xerampelina*) engaged in (A) a monandrous nest involving a single female and single male, (B) a polyandrous nest involving a single female and 11 males, (C) a communal nest involving 6–8 females and 30–40 males, and (D) a communal nest involving tens of females and hundreds of males. Photographs courtesy of Phillip Byrne.

through shared paternity? If males receive a fitness benefit from assisting with building polyandrous nests, we should expect that participating males will gain a share of paternity. We addressed these questions by studying a wild population for a complete breeding season and combining detailed observations of the construction of monandrous and polyandrous nests with measures of nest morphology, nest success, and paternity analyses.

Materials and methods

Study species and breeding biology

The African gray foam-nest treefrog, *C. xerampelina* is a large (snout-vent length [SVL]: 50–80 mm) prolonged breeding treefrog from the family Rhacophoridae that is widely distributed throughout southeastern Africa (Du Preez, 2017). The species has a prolonged breeding season, with reproduction typically occurring in the wet season between October and February (Du Preez, 2017). On average, females are approximately 20%–30% larger than males (Jennions et al., 1992) and lay clutches of 270–930 eggs in arboreal foam nests that typically take 4–6 hr to construct (Byrne & Whiting, 2008, 2011). The night after nest construction, females return to the nest site to add a final layer of foam (Jennions et al., 1992). Tadpoles develop in the nest, and after 5–6 days they break free from the foam and fall into water bodies below, where they metamorphose after approximately 6–8 weeks (unpublished data). Females can

mate with one male (monandrous matings; Figure 1A), multiple males (polyandrous matings; Figure 1B), or join the nest sites of other females (communal nesting) (Figure 1C, and Figure 1D). Polyandry has been reported in multiple populations (Byrne & Whiting, 2008; Jennions et al., 1992), with evidence that 66% of nesting events involve multiple males (average = 5.5 males) (Jennions et al., 1992).

Study population

The study took place using a natural population of frogs located in Hans Merensky Nature Reserve (24°25'380S, 31°22'400E, 462 m a.s.l.) in Limpopo Province, South Africa. Frogs were breeding on tree branches overhanging a large (approximately 500 m²), semi-permanent water body (Tsonga Kraal Dam) that was fed by an ephemeral stream and surrounded by native savanna bushveld dominated by mopane trees, *Colophospermum mopane* (Figure 2). On November 18, 2006, the dam filled following a significant rain event. On November 19, males entered the site and started calling, and on November 20, females entered the site and started breeding. We collected data on 99 consecutive nights between sundown (approximately 18:00 hr) and sunrise (approximately 07:00 hr) over an entire breeding season, with the last night of breeding activity recorded on March 3, 2007. Throughout the study period, ambient temperature ranged between 19 and 28.5°C.



Figure 2. Breeding site used by the African foam-nest treefrog (*Chiromantis xerampelina*) at Hans Merensky Nature reserve in Limpopo Province, South Africa. Images depict (A) several foam nests constructed on tree branches overhanging water in the Tsonga Kraal dam and (B) frogs aggregating on tree branches at the start of a breeding night. Photographs courtesy of Phillip Byrne.

Observations of nesting behavior

To assess the degree of synchronization of timing (start and finish) in bouts of leg-churning behavior between spawning frogs, throughout the breeding season we opportunistically video recorded bouts of female leg churning during nest construction under infrared light using a SONY handheld video camera (model DCR-HC42) set on night-shot mode. All recordings were made at a minimum distance of ~4 m to reduce disturbance to mating frogs. We only filmed nests involving 1–6 males because larger spawnings were too difficult to accurately observe the behavior of individuals. From these recordings, we randomly selected 65 female churning bouts (ranging in duration from 5 to 26 s) from 65 independent nests (i.e., a single churning bout per nest) where the number of frogs involved remained constant during nest construction (i.e., no males left or joined the nest). During nest construction, females occasionally churned their legs very briefly (approximately 1 s in duration) and stopped. These movements were not deemed to be proper churning bouts and were ignored in later behavioral analysis (see data analysis section for details on how synchrony in bouts of leg churning was assessed). When describing behavior, we refer to the male grasping the female as the “amplectant male” and all other males involved in the spawning as “non-amplectant males”. In addition to the video recordings, we observed the formation and construction of 134 nests involving between 1 and 12 males. All nest constructions were observed from start to finish so that the number of males could be accurately quantified and related to nest size (see below). Observations of nest construction took place on 36 nights, and during these nights it was possible to monitor up to 14 nest constructions per night because discrete nests were often constructed within several feet of each other. Frogs were toe-clipped when first caught to allow identification of previously observed frogs and ensure independence of our nesting observations.

Nest measurement and quantification of nest success

All nests ($n = 134$) were measured approximately 24 hr after nest construction to allow females to revisit the nest the fol-

lowing night to add a layer of foam (Jennions et al., 1992). For each nest, we measured height, width, and depth to the substrate. Measurements were made to the nearest cm using a ruler and taken approximately 5 mm from the nest surface to minimize disturbance while ensuring accuracy. To reduce measurement error, we measured nests three times and calculated an average. We checked nests every afternoon between 16:00 and 18:00 hr for eight consecutive days and categorized them as successful (nest structure was retained and tadpoles successfully hatched and left the nest) or unsuccessful (nest structure failed and no tadpoles left the nest). When nests failed, the reason for failure was recorded. Nest volume was estimated using the equation for an ellipsoid: volume (mls) = $(4/3 \pi abc)$ where a, b, and c are the semi-axes (i.e., half the length) of the measures of the nest height, nest width, and nest depth (all in cm) to the substratum that it was attached.

Tissue collection, genotyping, and paternity analysis

For paternity analysis, we observed six polyandrous nests (from the broader set of 134 nests) involving between 2 and 13 males, all of which were in bodily contact with the foam nest. All nest constructions were observed from start to finish between approximately 18:00 and 07:00 hr, with nests taking approximately 4–6 hr to build. This extended duration of nesting made it logistically difficult to accurately observe complete nesting events, explaining the relatively small final sample size. Nests were deemed to have been completed when frogs left the nest site. At this point, all frogs were collected and individually contained in ventilated plastic zip-lock bags labeled to identify an individual's sex. For all males involved, we also recorded whether they were the “amplectant male” or a “non-amplectant male”. In order to relate amplectant position to paternity success, we only considered nests where the same male remained as the amplectant male for the entire nest construction event (i.e., there was no switching of amplectant males when females left the nest to rehydrate). For each frog, we measured SVL to the nearest millimeter using a plastic ruler, and we used surgical scissors (sterilized in ethanol) to take a toe clip that was immediately

stored in an Eppendorf tube containing 95% ethanol. No frogs were held for longer than 15 min and all frogs were returned to the point of capture. Nests were collected the day after spawning and moved to a temperature-controlled room (maintained between 26 and 31°C) in the Tsonga Krall Museum (located ca. 20 m from the dam), where they were enclosed in an aluminum wire mesh frame and suspended within a 10-L plastic bucket containing 5 L of fresh dam water. Each bucket was enclosed with a plastic bag to prevent nest desiccation, and after ~72 hr, the nests were gently agitated to release tadpoles from the congealed foam. We then used a nylon fish net to catch and count every tadpole in a clutch before randomly sampling approximately 15%–20% of individuals that were then euthanized and stored in specimen vials containing 95% ethanol. Remaining tadpoles were released into the dam. Of note, there were no dead tadpoles observed in any nests, indicating that moving the nests did not compromise tadpole survival.

Adults and offspring were genotyped using a large SNP data set generated by Diversity Arrays Technology using the DArTseqTM approach, which represents a combination of DArT complexity reduction methods and next-generation sequencing platforms (Kilian et al., 2012). The background and process for determining paternity have been outlined in detail elsewhere (Bookmythe et al., 2016; Head et al., 2017; Vinogradov et al., 2025). The data set contained approximately 8,530 SNPs with an average call rate of 90.0% and a reproducibility of 99.5%. Using these SNPs, we calculated a Hamming Distance Matrix (HDM) of all successfully genotyped individuals to determine parentage (Hu et al., 2015). The HD is a measure of genetic dissimilarity across the full SNP dataset. All offspring in a clutch were lined up against each other as well as the dam and putative sires and compared to both confirm relatedness to the captured dam and to determine paternity from the captured putative sires. For our data set, HD values for parents and offspring ranged from approximately 0.05 to 0.07, whereas unrelated animals had HD values that ranged from approximately 0.09 to 0.14. Paternity could be unambiguously assigned when the putative father was sampled, but in some cases the sires of some offspring were not captured.

Data analysis

To assess the degree to which nesting frogs synchronize the timing of bouts of leg-churning behavior, we scored whether each male involved in 65 independent female leg-churning bouts was or wasn't displaying leg movement. This was done for each second of the female leg-churning bout, as well as for each second of a preceding control period that was matched in duration to that of the leg-churning bout. To calculate whether leg-churning behavior between spawning frogs is more synchronous than expected by chance, we calculated the Kappa coefficient of agreement (K) (Asher & Collins, 2012). Synchrony was calculated between pairs of frogs (males and females) using Cohen's Kappa, and between more than two frogs (the average pairwise agreement across all individuals at a nest) using Fleiss' Kappa. The Kappa coefficient is the preferred measure of behavioral synchrony as it controls for expected levels of synchrony due to chance as well as for effects of group size, which can artificially inflate the percentage agreement (Asher & Collins, 2012; van Dixhoorn et al., 2024). A Kappa value

(which ranges between -1 and $+1$) expresses the degree to which the observed agreement in behavior (e.g., churning or not churning) between individuals at a specific time point exceeds the level of agreement expected if all individuals were behaving randomly. Negative K values indicate that inter-individual agreement is less than the agreement expected by chance, with zero values indicating that agreement is no better than chance. Increasingly positive K values represent increasing "better-than-chance" agreement for the observed individuals, with $+1$ indicating perfect agreement. Kappa values were calculated using DATAtab (Numiqo: Online Statistics Calculator. DATAtab e.U. Graz, Austria. URL <https://numiqo.com>). To test if the Kappa values were significantly different from zero (i.e., the probability that the observed agreement is not due to chance), we used Z-tests. Generalized regression analysis with a logit function and beta distribution showed no significant relationship between the duration of a female churning bout and the degree of synchrony (Kappa coefficient) for either entire group of nesting frogs (generalized regression analysis; generalized $R^2 = 0.005$, $\beta = 0.02$, $SE = 0.03$, Wald $\chi^2 = 0.34$, $p = 0.55$) or amplexant pairs alone (generalized regression analysis; generalized $R^2 = 0.001$, $\beta = 0.01$, $SE = 0.03$, Wald $\chi^2 = 0.08$, $p = 0.77$). Therefore, duration of the churning bout was not included in subsequent analyses. Generalized regression analysis with logit function and beta distribution was used to assess the relationship between the number of males involved in a nest and the degree of synchrony for the entire group of nesting frogs, as well as the amplexant pair alone. Generalized regression analysis with logit function and beta distribution was also used to assess the relationship between the number of males involved in a mating and the proportion of time that female leg churning bouts involved male-leg churning (from any male). A generalized linear mixed model (GLMM) with logit function and beta distribution was used to assess whether there was a difference between monandrous and polyandrous nests in the proportion of time of a female churning bout that individual males churned. In the model, nest type (monandrous vs. polyandrous) was the fixed effect, female ID was the random effect, and proportion of male churn time was the response variable. Because the proportion data used in the generalized regression analyses and the GLMM contained boundary values (zeros and ones), prior to analysis the data was transformed (compressed) using the equation $y' = \frac{y(n-1)+0.5}{n}$, where y was the proportion data, and n was the sample size (following Smithson & Verkuilen, 2006). Because female SVL has previously been found to positively correlate with nest size in a foam-nesting anuran, presumably due to larger females having an increased ability to release a higher volume of water and oviducal secretion needed for nest construction or having greater physical ability to mechanically agitate the foam (Kasuya et al., 1987), simple regression analysis was used to assess the relationship between female SVL and nest volume (mls) in monandrous and polyandrous nests. Simple regression analysis was also used to assess the relationship between the number of males involved in nest construction and nest volume (mls). To assess the relationship between the number of males involved in nest construction (1–12) and nest viability (failed or succeeded), we used logistic regression analysis. While a small number of nests ($n = 4$) failed because they blew down in heavy wind, all

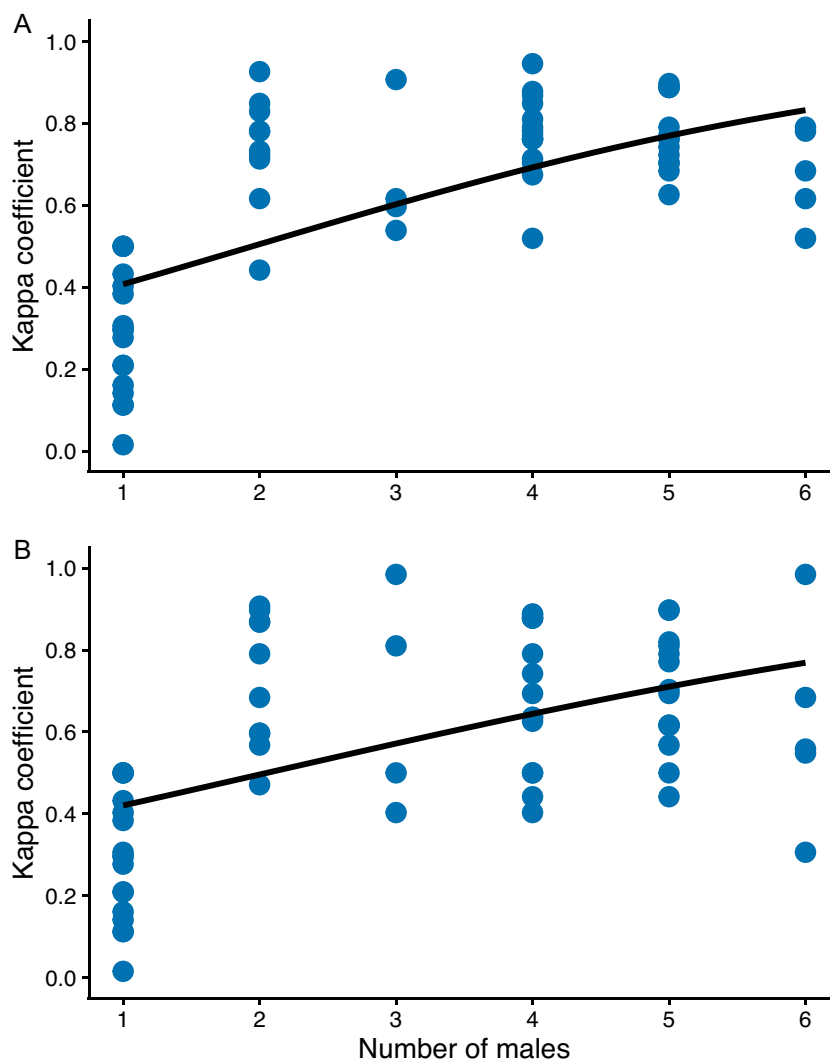


Figure 3. Output of a generalized regression analysis with beta distribution (including a fitted regression curve) showing the relationship between the number of males involved in nest construction in the African foam-nest treefrog (*Chiromantis xerampelina*) and the degree of synchrony in leg churning (calculated as the Kappa coefficient) for (A) all of the frogs involved in the nest construction, and (B) the amplexant pair involved in nest construction. Kappa values were transformed to handle boundary values (0 s and 1 s).

other nest failures were caused by complete nest desiccation. Therefore, to assess the relationship between the number of males involved in nest construction and nest failure caused by desiccation, we repeated the logistic-regression analysis excluding the nests that failed because they blew down in wind. Analyses were performed in JMP Pro version 18.

Ethics permission

The study was conducted with the permission of the University of Witwatersrand Ethics Committee (#2007).

Results

Do frogs synchronize leg-churning bouts during nest construction?

Across 65 independent female churning bouts (representing 65 different nests), Kappa-coefficient values were much higher for polyandrous nests than monandrous nests

(Supplementary Table S1). Kappa-coefficient values for nests involving a female and a single male ranged between 0 and 0.5, while Kappa-coefficient values for nests involving a female and multiple males (2–6) ranged between 0.44 and 0.96 (Supplementary Table S1). These values show that spawning frogs displayed a low to moderate level of agreement (synchrony) in leg-churning bouts in monandrous nests, and a substantial to near-perfect level of agreement in polyandrous nests. Across all matings, Z-tests revealed that the observed level of agreement was only significantly different from zero for 23.52% (4/17) of monandrous nests but was significantly different from zero for 100% (48/48) of polyandrous nests (Supplementary Table S1). Generalized regression analysis showed that there was a significant positive association between the number of males involved in nest construction and the degree of synchrony (Kappa coefficient) (generalized $R^2 = 0.36$, $\beta = 0.47$, $SE = 0.08$, Wald $\chi^2 = 33.79$, $p < 0.001$; Figure 3A).

When considering the level of agreement in leg-churning bouts for amplexant pairs only (including those in polyandrous nests), Kappa-coefficient values ranged between 0

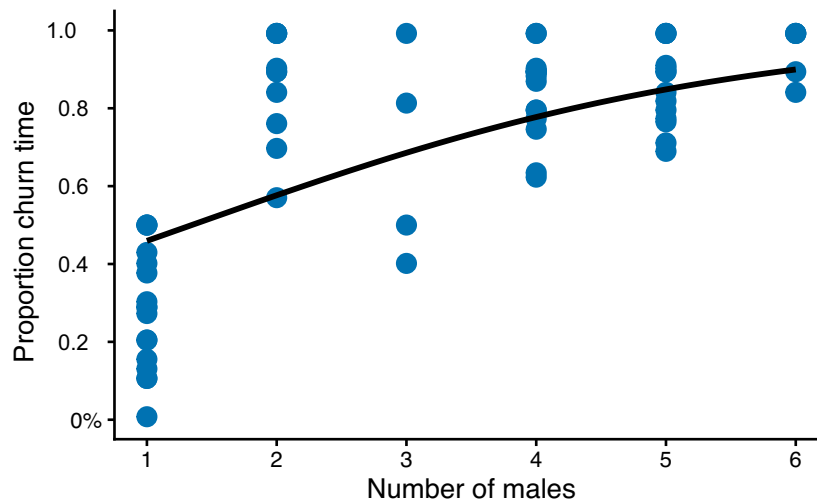


Figure 4. Output of a generalized regression analysis with beta distribution (including a fitted regression curve) showing the relationship between the number of males involved in nest construction and the percentage of a female's leg churning bout accompanied by male leg churning during nest construction in the African foam-nest treefrog (*Chiromantis xerampelina*).

and 1.0, showing that there was a highly variable level of agreement between pairs in their leg-churning bouts (Supplementary Table S2). In monandrous nests, agreement was low to moderate (0–0.5), and Z-tests revealed that the observed level of agreement was not significantly different from zero in 76.47% (13/17) of nests. In polyandrous nests, agreement was moderate to substantial (0.3–1.0), and Z-tests revealed that the observed level of agreement was significantly different from zero in 97.91% (47/48) of nests (Supplementary Table S2). Generalized regression analysis showed that there was a significant positive association between the number of males involved in nest construction and the degree of synchrony (Kappa coefficient) (generalized $R^2 = 0.24$, $\beta = 0.31$, $SE = 0.07$, Wald $\chi^2 = 19.87$, $p < 0.001$; Figure 3B).

Do nests involving more males have a higher proportion of male-leg churning activity during nest-building bouts?

The number of males involved in nest construction was significantly positively related to the proportion of time that any male was observed leg churning during a female leg-churning bout (generalized regression analysis; generalized $R^2 = 0.36$, $\beta = 0.47$, $SE = 0.08$, Wald $\chi^2 = 33.79$, $p < 0.001$; Figure 3). In nests involving one male, less than 20% of a female's leg-churning bout involved male leg churning behavior. In nests involving 2–6 males, between approximately 30% and 95% of a female's leg-churning bout also involved at least one male displaying leg churning behavior (Figure 4). The leg movements of amplexant and non-amplexant males in polyandrous nest constructions were much more pronounced and vigorous than amplexant males in monandrous nest constructions (see Supplementary Videos S1 and S2). Within nests, individual males also displayed a significantly higher proportion of leg churning when involved in the construction of polyandrous nests (range = 0.2–1.0) compared to monandrous nests (range = 0–0.5; GLMM: $F_{1,68} = 50.21$, $p = 0.0001$).

Are nests with more males larger and less susceptible to failure?

For matings involving between 1 and 12 males, nest volume ranged between 47.12 and 1319.47 ml (mean $SEM = 476.04 \pm 22.10$, $n = 134$ nests). Nest volume and female SVL were positively and significantly correlated in monandrous nests, although this relationship was weak, with female size only explaining 25% of the variation in nest volume (simple regression analysis: $R^2 = 0.25$, $F_{1,53} = 17.36$, $p < 0.0001$). For polyandrous nests, nest volume and female SVL were not significantly correlated, with female SVL explaining only ca. 3% of the variance in nest volume (simple regression: $R^2 = 0.03$, $F_{1,77} = 2.14$, $p = 0.147$). The number of males involved in the construction of a nest and nest volume was positively and highly significantly correlated, with the number of males explaining more than 70% of the variation in nest volume (simple regression: $R^2 = 0.71$, $F_{1,132} = 324.44$, $p < 0.0001$, Figure 5). Polyandrous nests (involving up to 12 males) were nearly three times the volume of the largest monandrous nests (Figure 5). Of the nests monitored, 30.55% (44/134) failed before tadpoles hatched. Of the failed nests, 9.09% (4/44) blew down during strong wind. Half of these nests were smaller monandrous nests (268.08 ml and 381.70 ml), while the other half were smaller to moderately sized nests (362.85 ml and 403.17 ml) constructed by a female and either 4 males or 5 males (for nest size distribution, see Figure 5). Of the remaining nests that failed, 29.85% (40/134) completely desiccated before tadpoles hatched (Figure 6). We classified nests as completely desiccated through visual inspection and a touch test. Desiccated nests were visibly shriveled with no moisture left inside, and displayed a texture that was hard and brittle, compared with the soft, spongy, and sticky texture of hydrated viable nests. When touched, desiccated nests quickly broke into powdery pieces, compared to hydrated nests, which retained their structure. Within desiccated nests, desiccated eggs and embryos were shriveled and yellow in appearance (for images, see Figure 6B). Of the desiccated nests, 82.5% (33/40) were monandrous nests toward the smaller end of the size distribution (approximate volume

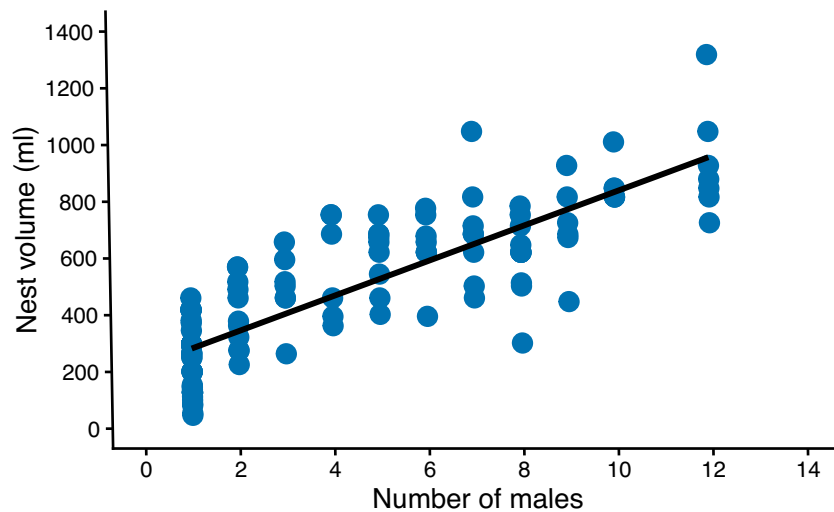


Figure 5. Correlation between the number of males involved in nest construction and foam-nest volume (ml) in the African foam-nest treefrog (*Chiromantis xerampelina*).

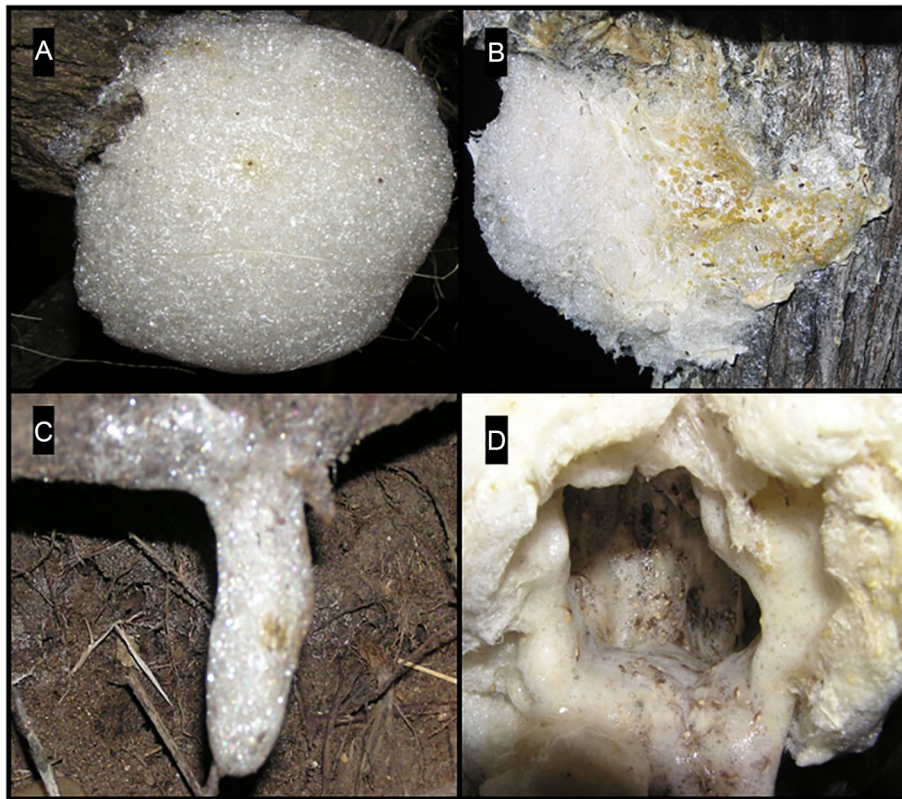


Figure 6. Images showing foam nests in the African foam-nest treefrog (*Chiromantis xerampelina*) in different stages of desiccation. Images depict: (A) a non-desiccated nest produced from a monandrous mating 1 day after construction, (B) a partially desiccated nest from a monandrous mating 2 days after construction, (C) a fully desiccated nest from a monandrous mating 4 days after construction, (D) a successful nest from a polyandrous mating 5 days after nest construction, with tadpoles breaking free from the foam and dropping into the water below. Photographs courtesy of Phillip Byrne.

= 54.97–296.88 ml). The remaining 17.5% of desiccated nests (7/40) were more moderately sized polyandrous nests (approximate volume = 596.12–753.98 ml) involving 2–7 males. The number of males involved in the construction of a nest had a significant positive effect on the likelihood that a nest succeeded (logistic regression: $\chi^2 = 40.94$, $d.f. = 1$,

$p < 0.0001$; Figure 7). Excluding nests that failed because they blew down during strong wind ($n = 4$), the number of males involved in nest construction had a significant positive effect on the likelihood that a nest avoided complete desiccation before tadpoles hatched (logistic regression: $\chi^2 = 40.99$, $d.f. = 1$, $p < 0.0001$).

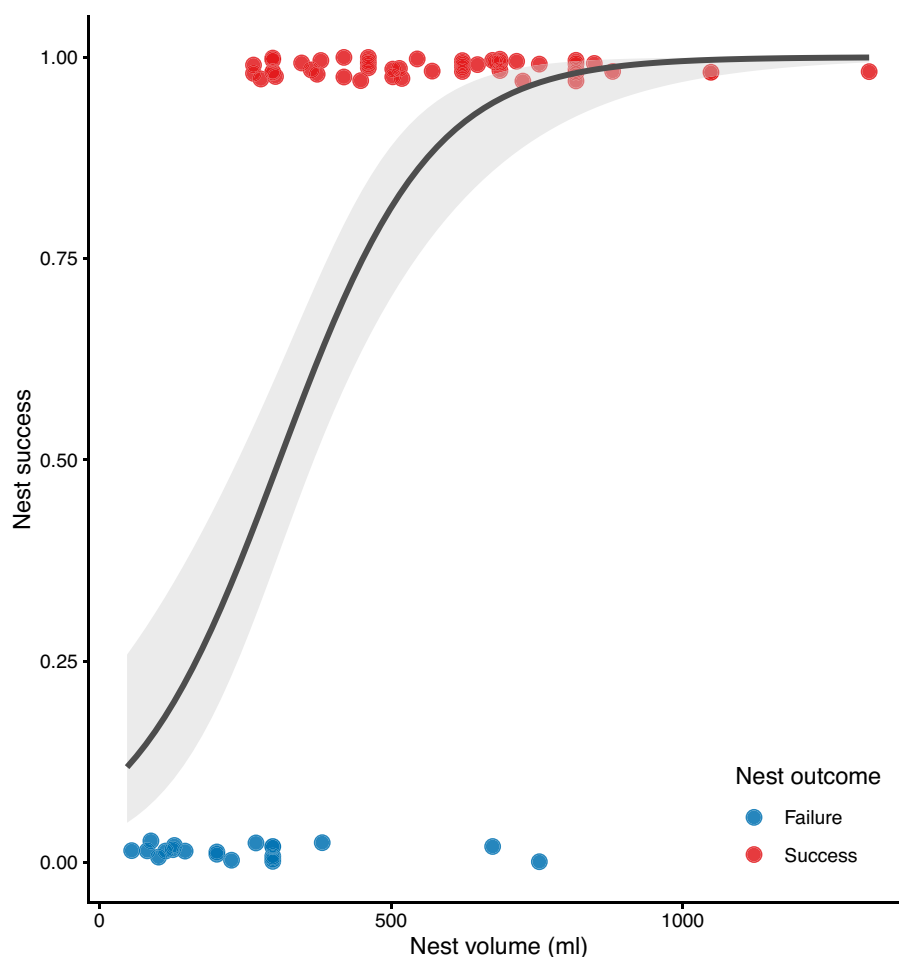


Figure 7. Output of a logistic-regression analysis showing the relationship between nest success and foam-nest volume (ml) in the African foam-nest treefrog (*Chiromantis xerampelina*).

Do males involved in polyandrous nests share paternity?

Multiple paternity was evident in 100% (6/6) of the polyandrous nests examined, yet the patterns of paternity within matings were highly variable (Table 1). In nest 1 ($n = 2$ males), paternity was approximately evenly split between the amplexant male and the non-amplexant male. In nest 2 ($n = 7$ males), paternity was divided between 6 males, with two non-amplexant males collectively siring 91% of the offspring. In nest 3 ($n = 8$ males), paternity was divided between 6 males, with the amplexant male and two non-amplexant males collectively siring 80.3% of the offspring. In nest 4 ($n = 10$ males), paternity was divided between just 4 males, with the amplexant male failing to sire any offspring, and two non-amplexant males collectively siring approximately 94% of the offspring. In nest 5, ($n = 12$ males), the amplexant male failed to sire any offspring, and paternity could be definitively assigned to at least 7 non-amplexant males. However, because 5 males were not caught and genotyped, there were 36 of the 160 offspring sampled that could not be assigned to a sire. As such, it is possible that up to 11 males gained a share of paternity. In nest 6 ($n = 13$ males), paternity was divided among 11 males, including the amplexant male and 9 non-amplexant males, with the most successful male siring 32.7% of the offspring (see Table 1).

Discussion

While numerous material benefits of polyandry have been demonstrated, there have been limited attempts to link the adaptive value of this mating system to nesting behavior. The aim of our study was to introduce the “nesting-assistance” hypothesis for the evolution of polyandry and provide an initial examination of this hypothesis in the African gray foam-nest treefrog. By studying a wild population over an entire breeding season, we (i) elucidated the extent to which males and females synchronize bouts of a conspicuous reproductive behavior (leg-churning) essential for foam-nest construction; (ii) showed positive relationships between the number of males involved in matings and nest size and resistance to desiccation; and (iii) highlighted the potential for multiple paternity to elevate fitness and selectively favor male assistance with nest building. Overall, we provide evidence that by mating with multiple males, females produce more viable nests that improve the survival of their offspring prior to hatching.

Past reports on the nesting behavior of African foam-nest treefrogs have inferred that males are assisting females to build nests through the simultaneous and rhythmic movement of hind limbs (Coe, 1967). Yet, there has been no attempt to quantify the level of synchrony in leg-churning bouts, or the proportion of time that males churn relative to

Table 1. Percentage of offspring sired by male African foam-nest treefrogs (*Chiromantis xerampelina*) participating in six polyandrous nests. Male 1 is the amplexant male, while males 2–13 are non-amplexant males. The numbers in brackets represent the number of offspring sired out of the total number sampled per clutch (totals = number sampled/clutch size). For each nest, approximately 25% of the clutch was sampled.

Male number	Nest 1	Nest 2	Nest 3	Nest 4	Nest 5	Nest 6
1	58.47% (69)	2.5% (5)	30.05% (61)	0% (0)	0% (0)	8.3% (23)
2	41.53% (49)	4.5% (9)	10.84% (22)	68.57% (48)	23.75% (38)	6.83% (19)
3		57.5% (115)	4.43% (9)	0% (0)	0.625% (1)	0% (0)
4		33.5% (67)	4.43% (9)	0% (0)	23.125% (37)	6.11% (17)
5		1% (2)	0% (0)	0% (0)	1.25% (2)	32.73% (91)
6		0% (0)	31.53% (64)	0% (0)	11.875% (19)	15.10% (42)
7		1% (2)	0% (0)	0% (0)	16.875% (27)	7.55% (21)
8			18.72% (38)	25.71% (18)	? (0–36)	1.08% (3)
9				4.29% (3)	? (0–36)	0.36% (1)
10				1.43% (1)	? (0–36)	0% (0)
11					? (0–36)	12.59% (35)
12					? (0–36)	1.8% (5)
13						7.55% (21)
Totals	118/790	200/1,012	203/1,100	70/466	160/800	278/1,340

Note. In nest 5, there were 5 males (males 8–12) that were not captured and genotyped. One or more of these males were responsible for siring 36 of the 160 offspring sampled.

females. Here, we demonstrate for the first time that there is a high level of synchrony in the timing of bouts of leg churning during nest construction, with distinct periods of inactivity punctuated by short bursts of hind limb movement by both sexes. In addition, we provide evidence that the degree of synchrony is much higher during the construction of polyandrous nests. This difference was the outcome of amplexant males involved in monandrous nests not always moving their hind legs during female-churning bouts. By comparison, both amplexant and non-amplexant males involved in polyandrous nest construction consistently displayed hind-leg movement during churning bouts. The likely reason for this difference is that males involved in polyandrous nests are stimulated to move more, not just by interactions with the female but also by those with neighboring males. An explanation supported by our demonstration of a positive relationship between the number of males building a nest and the proportion of time allocated to leg churning by individual males. Studies of behavioral synchronization have previously shown that animals in moving groups adjust their movements in response to the movements and positions of their neighbors (termed interaction rules), with these interactions resulting in coordinated group movements (Herbert-Read, 2016). As an outcome of the elevated leg movement, the proportion of a bout of churning that involved at least one male churning at the same time as the female was significantly higher in polyandrous nests. While it remains unclear whether male leg churning in *C. xerampelina* has evolved specifically for the purpose of nest building or has its origins in benefits linked to other primary functions (e.g., the distribution of sperm over eggs) (Jennions et al., 1992; Kasuya et al., 1996), the heightened level of leg movement we observed in polyandrous mating appears to have profound implications for nest morphology and viability.

As predicted, we observed a strong positive and highly significant relationship between nest size (measured as volume) and the number of males involved in nest construction. At the upper end of the nest-size distribution, polyandrous nests (involving up to 12 males) were nearly three times the volume of the largest monandrous nests. Fresh *C.*

xerampeliana foam is known to be comprised of 77% air (Seymour & Loveridge, 1994). Therefore, the most likely explanation for this relationship is that leg churning by more males increased the mechanical agitation of oviducal secretions and incorporated more air bubbles into the foam, increasing final nest size (Gould, 2021). While the biomechanical and chemical process through which air bubbles are integrated into anuran foam remains unclear, evidence that mechanical agitation is essential for nest formation comes from the observation that species without leg churning produce structurally flat nests (Hödl, 1997). Furthermore, in a polyandrous Japanese foam-nesting tree frog (*Rhacophorus arboreus*) where males do not display leg churning, no correlation has been found between nest size and number of mating males (Kasuya et al., 1987). However, it should be considered that male presence during nest construction might still have positive effects on nest size via other mechanisms. For instance, mucous excreted from male skin glands may contribute precursor fluid and proteins needed for nest building (Shahrudin et al., 2017). Another intriguing possibility is that components of male ejaculates (e.g., seminal fluid proteins) are used in foam production, as reported in the Indian Malabar Gliding Frog (*Rhacophorus malabaricus*) (Kadadevaru & Kanamadi, 2000). Beyond contributing substances that aid in foam production, males may also contribute chemicals that improve foam quality. For example, novel anuran proteins that act as surfactants (e.g., ranaspumins) are known to lower the surface tension of water and improve the integration of air bubbles into foam (Cooper et al., 2017). Moving forward, more research will be needed to elucidate the precise contributions that males make to nest construction. Important insights may come from biochemical characterization of the foam to ascertain the nature and source of compounds used in nest construction (Cooper et al., 2017). In combination, manipulative experiments that control or simulate male leg movement may help to elucidate agitation levels needed to optimize nest formation. To this end, it would also be insightful to experimentally test the extent to which synchrony in leg churning between nesting frogs influences nest architecture, both

independent of and in relation to the number of participating frogs. It is assumed that coordinated leg churning assists with the integration of more air into foam and generates larger nests, but this will need to be validated.

Regardless of the causal link between male presence and increased nest size, a critical finding was that larger nests from polyandrous matings were less likely to desiccate. The strength of this association was profound. Over the breeding season 30% of nests failed, and 82.5% of these were smaller monandrous nests. The other 17.5% were from more moderately sized nests produced by matings involving 2–7 males. No larger nests resulting from polyandrous matings failed due to desiccation. The simplest explanation for this outcome is that larger nests were less susceptible to water loss and more resistant to desiccation. A fundamental geometric relationship is that larger spheres/ellipsoids have reduced surface area to volume ratios. Knowing this, we should expect that exposure of a smaller nest-surface area to dry air will reduce the rate of evaporative water loss and lower the risk of egg desiccation (Gould, 2021). Larger nests that contain more air bubbles and increased structural complexity may also provide a more effective physical barrier to water loss (Gould, 2021). While relationships between anuran foam-nest size and water loss are yet to be rigorously investigated, there is some evidence that increased size elevates desiccation resistance. Seymour and Loveridge (1994) held *C. xerampelina* nests in the lab under conditions of constant temperature and humidity and reported that relatively small foam nests suffered a higher percentage of desiccated eggs (Seymour & Loveridge, 1994). More recently, Zina (2006) held Tungara frog (*Physalaemus pustulosus*) foam nests under completely dry conditions and reported that larger nests (produced through communal spawning) had significantly lower relative rates of water loss (Zina, 2006). Beyond effects on nest size, polyandry may also influence desiccation resistance through impacts on foam quality. Frog foam is known to contain a cocktail of unique proteins and carbohydrates (including powerful surfactants) that have anti-desiccation properties (Cooper et al., 2017). As mentioned above, if males contribute unique compounds to the foam through their skin or ejaculate, the potential exists for polyandrous females to not just make larger nests, but those that have a higher level of chemical protection against water loss.

For males, participation in polyandrous nest construction has significant fitness implications. Using a genetic approach to analyse parentage, we unambiguously showed that males stand to gain a share of paternity when spawning alongside other males. There was, however, considerable variation between polyandrous nests in how paternity was divided. In some nests, paternity was split relatively evenly, while in others it was biased toward a subset of participants, with some males failing to sire offspring altogether. There was no evidence that paternity success was related to whether a male was in the amplexant or non-amplexant spawning position. Remarkably, in two of the six nests we examined, amplexant males failed to sire any offspring. We speculate that variation in paternity success may have arisen because some males either refrained from releasing sperm, released too few sperm, released inviable sperm, or released sperm of low competitive ability. This supposition aligns with a previous study where we reported low fertilization success in monandrous nests (Byrne & Whiting, 2008). Say-

ing this, we must interpret the paternity data with caution because our sampling of each clutch was incomplete. For ethical reasons, we were restricted to randomly sampling approximately 15%–20% of the offspring from each clutch. More comprehensive sampling within clutches, combined with detailed behavioral analysis, will be needed to ascertain exactly how paternity is shared between mating males, and how variation might relate to male phenotype, physiological state, or behavior during spawning. Nevertheless, by demonstrating multiple paternity, we can conclude that males stand to gain fertilizations by participating in the construction of polyandrous nests. Knowing that polyandrous matings have a higher probability of producing viable nests with higher offspring survival to hatching, ensuring that polyandrous nests are well built should be mutually beneficial to all male participants. If true, we should expect strong selection against any form of male agonistic or competitive behavior that interferes with nesting outcomes, as is commonplace in mammals and birds that display cooperative polyandry (Cao et al., 2010; Goldizen, 1989). Indeed, we saw no evidence for physical or acoustic aggression, as observed in other *C. xerampelina* populations (Jennions et al., 1992). Instead, once a spawning commenced, males typically remained in the same mating position and only displayed minor jostling as they moved to juxtapose their cloaca with the females during egg release. Males also directed no behaviors toward females typically associated with sexual coercion in anurans, such as harassment, repeated amplexant attempts or multiple-male amplexus (Byrne & Roberts, 1999). The possibility should be considered that there is some level of cooperation between nesting frogs. However, we emphasize that it is equally possible that high costs of aggression have favored male restraint, with males simply behaving in a way that enhances their own fertilization success and fitness, irrespective of fitness impacts on other males or females. Support for this idea comes from our observation that males move their legs more during nest construction when other males are present, possibly to help distribute sperm through foam and improve fertilization success under sperm competition. Given this possibility, it remains unclear whether males churn their legs to directly assist in polyandrous nest construction, or whether assistance is a side effect of physical interactions with other males. Elucidating the selective pressures favoring male leg churning during nest construction will require a combination of targeted field observations and behavioral experiments designed to identify the primary function of this behavior, and this should be a priority for future research.

Taken together, our results provide support for the “nesting-assistance” hypothesis for the evolution of polyandry. The increase in nest viability and reproductive success we report suggests a substantial and direct fitness benefit to females of simultaneously mating with multiple males. Previously, we have reported that polyandry in *C. xerampelina* can improve female fitness by safeguarding against male infertility (Byrne & Whiting, 2008) and improving the genetic diversity and viability of offspring post-hatching (Byrne & Whiting, 2011). The findings of our present study indicate that mitigating desiccation risk to nests and offspring is an additional reproductive benefit favoring the evolution of polyandry. Our argument that nesting assistance is critical for nest viability and the fitness of both sexes is strengthened by knowledge that female

C. xerampelina also engage in communal spawning. These spawnings can involve tens of females and hundreds of males and produce enormous nests. We monitored 42 nests from communal spawns, and none of them failed due to desiccation (unpublished data). Given this benefit, a logical question to ask is why communal spawning has not replaced polyandry as the dominant reproductive strategy? The likely answer is that female reproductive behavior is tightly linked to fluctuating climatic conditions and is highly dependent on frog density and the OSR. Communal spawning was mostly observed during periods of extremely high density (with less male-biased OSRs) on nights when conditions were ideal for breeding (i.e., warm nights with low wind and high humidity after heavy rain; unpublished data). On these nights, female arrival was highly synchronized (making the OSR less heavily male-biased), and large numbers of males were present at preferred nesting locations. On nights where the density of breeding frogs was lower and female arrival was more asynchronous (and the OSR was subsequently more heavily male-biased), the opportunity for group spawning was reduced and polyandrous spawning was more common. Given these breeding dynamics, to better understand the evolution of the *C. xerampelina* mating system, it will be necessary to investigate factors influencing female-nesting decisions, with an emphasis on better understanding how females maximize their chances of mating with multiple males on any given night. It may be the case that females preferentially nest on tree branches with higher male density (Jennions et al., 1992) or that females use visual, auditory, or odor cues to exaggerate their presence and elicit male participation (Akre & Ryan, 2011; Byrne & Keogh, 2007; Chen et al., 2024). In parallel, in order to evaluate the net fitness gains of polyandry, it will be necessary to quantify if females incur any costs from mating with multiple males (e.g., greater energy expenditure or exposure to predators or disease) (Gowaty, 2012). Quantifying exactly how female and male reproductive behavior changes with fluctuations in density and the OSR, and measuring potential mating costs, will be important next steps toward understanding the factors shaping the *C. xerampelina* mating system and foam-nesting polyandrous mating systems more broadly.

Among other foam-nesting anuran amphibians, fitness gains linked to male-nesting assistance and the protection of offspring against nest desiccation might help to explain the evolution of polyandry in a diversity of species. Foam nesting is widespread among frogs, with the involvement of multiple males in nest construction observed in at least 18 species across 7 genera: *Chiromantis* (Coe, 1967, 1974; Jennions et al., 1992), *Eupemphix* (Giarretta & Facure, 2006), *Leptodactylus* (Prado & Haddad, 2003), *Physalaemus* (Barreto & Andrade, 1995; Giarretta & Menin, 2004; Hödl, 1990; Villamil & Maneyro, 2014; Zina, 2006), *Pleurodema* (Cardoso & Arzabe, 1993; Hödl, 1992), *Polypedates* (Feng & Narins, 1991; Meegaskumbura et al., 2011), and *Rhacophorus* (Fukuyama, 1991; Kasuya et al., 1987; Liao & Lu, 2010). It is noteworthy that for some of these species polyandrous matings may arise due to selfish male behavior, with sneaker males joining pairs and stealing fertilizations. This certainly appears to be the case for the Japanese treefrog, *R. arboreus*, where males fight to amplex females and display almost no leg movement during nest construction (Kasuya et al., 1987; Kasuya et al., 1996). However, for most of the species where polyandry has been

observed, there appears to be some level of intra- and inter-sexual coordination during nesting, with males rhythmically churning legs and showing no or extremely limited aggression during spawning (Coe, 1967; Fukuyama, 1991; Hödl, 1992; Liao & Lu, 2010; Prado & Haddad, 2003). For some of these species, there is also evidence that nest desiccation can be a major cause of offspring mortality (Coe, 1974; Downie, 1988; Giarretta & Menin, 2004). Remarkably, in at least one species (the Chinese Omei tree frog, *Rhacophorus omeimontis*), the risk of nest desiccation appears to be so extreme that females have evolved to reduce the rate of water loss from nests by wrapping them in a protective layer of leaves (Liao & Lu, 2010). For species that face a high risk of nest desiccation, there would be value in critically evaluating the “nesting-assistance” hypothesis as a first step toward assessing the generality of our hypothesis. To this end, employing the comparative approach would also be insightful. Breeding in foam-nesting anurans occurs across a diversity of biomes, ranging from tropical rainforests to dry woodlands and semi-arid savannah, representing a distinct aridity cline. Since these habitats differ considerably in the evapo-transpiration rate, we presume they represent a distinct and broad spectrum of nest-desiccation risk. This inter-specific variation provides an excellent opportunity to undertake a phylogenetically controlled comparative investigation aimed at understanding the role that desiccation risk has played in the evolution of nesting assistance and polyandry. We predict that the frequency of polyandry, and the average number of males involved in matings, will be significantly higher in foam-nesting species from more arid environments that are challenged by a higher risk of nest desiccation. We also predict that a male-biased OSR will be a necessary condition for simultaneous polyandry and male-nesting assistance, so we encourage the collection of these data in future intraspecific studies.

More broadly, increased offspring survival resulting from male-nesting assistance and improved nest viability is a direct benefit of polyandry not previously reported. Therefore, our findings make an important contribution to our understanding of how material non-genetic benefits might shape female mating decisions and mating-system evolution. We argue that this direct benefit could favor polyandry in a spectacular diversity of animals. While the nesting-assistance hypothesis is unlikely to apply to cooperatively breeding mammals and birds where nesting assistance by subordinate helper males is strictly kin-selected (García-Ruiz et al., 2022), a review of the literature, focused explicitly on species where kin selection is likely to be of no or limited importance and multiple paternity has been demonstrated, indicates that male-nesting assistance takes various forms and is taxonomically widespread. Evidence of males helping to assist females with nest building, either directly through nest construction or indirectly by gathering materials or helping with nest location, can be found in New World monkeys (e.g., tamarins and marmosets: Goldizen, 1988; Smith et al., 2007), canids (e.g., arctic foxes and gray wolves: Ausband, 2018; Carmichael et al., 2007), raptors (e.g., Harris hawks and Galapagos hawks: DeLay et al., 1996; Faaborg et al., 1995; Mader, 1979), seabirds (e.g., common terns and Red-footed Booby: Cao et al., 2010; Ludwigs, 2004), waterfowl (e.g., dusky moorhens and Tasmanian native hens: Garnett, 1980; Gibbs et al., 1994; Goldizen et al., 1998; Putland & Goldizen, 2001), sediment-spawning cichlid fishes (e.g.,

arapaima fish and cave-spawning cichlid fish: Awata et al., 2005; Farias et al., 2015), monitor lizards (e.g., Rosenberg's goanna: Rismiller et al., 2010), leaf-folding anuran amphibians (Pyburn, 1970; Wogel et al., 2005), solitary bees (e.g., carpenter bees: Mikát et al., 2019, 2021), and solitary wasps (e.g., organ-pipe mud-dauber wasps: Brockmann, 1992). In these species, nests are known to play a critical role in protecting offspring from various biotic and/or abiotic challenges (e.g., predation, parasitism, and severe weather) during early development. Surprisingly, there is a complete absence of data on the impact of male nesting assistance on nest viability and offspring survival. Moving forward, we forecast that investigation into the impacts of male assistance on nesting success in a diversity of species will provide additional support for the "nesting-assistance" hypothesis. Such studies are strongly encouraged to help elucidate how behaviors that mitigate ecological threats to offspring survival can contribute to the evolution of polyandrous mating systems.

Supplementary material

Supplementary material is available online at [Evolution](#).

Data availability

The SNP data, and the Hamming Distance Values generated from the SNP data, is available from the Dryad repository (DOI: [10.5061/dryad.9s4mw6mx8](#)).

Author contributions

P.B. conceived the study with input from M.W. P.B. collected the field data. J.K. ran and interpreted the genomic analyses. P.B. analyzed the field data and wrote the manuscript, with input from J.K., M.W., and A.S.

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Conflict of interest

The authors declare no conflicts of interest.

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